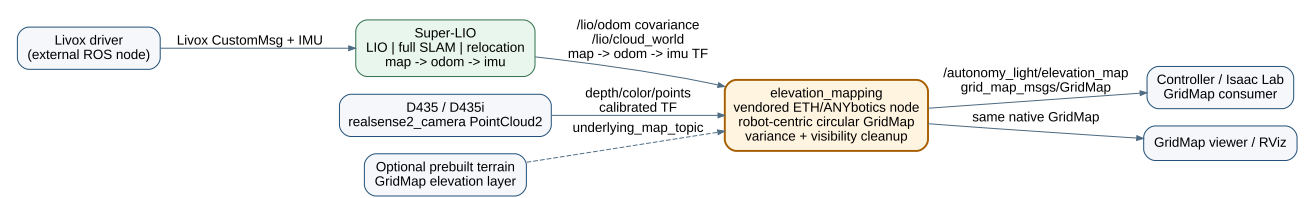


# autonomy\_light software architecture

Super-LIO + ETH/ANYbotics robot-centric elevation mapping · 2026-08-04

**Design decision.** This repository no longer contains an elevation-map algorithm, custom message, DDS bridge, or D435 merge node. It brings up one vendored ETH/ANYbotics elevation mapping node; Super-LIO remains the single owner of LIO, full SLAM, relocation, and `map → odom`.



## Node responsibilities

| Node                           | Owner                  | Responsibility                                                                                                              |
|--------------------------------|------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Livox driver                   | external               | Publishes one Livox CustomMsg and IMU stream.                                                                               |
| Super-LIO                      | external               | LiDAR–IMU odometry, ESKF covariance, full SLAM, saved-map relocation, and <code>map → odom → imu</code> .                   |
| D435/D435i driver              | external               | Publishes one PointCloud2 stream with its calibrated TF chain.                                                              |
| <code>elevation_mapping</code> | vendored ETH/ANYbotics | Robot-centric circular grid, sensor-specific variance, pose-covariance prediction, Bayesian fusion, and visibility cleanup. |
| <code>autonomy_light</code>    | this repository        | Configuration, process lifecycle, and calibrated static TF only.                                                            |

## Data and TF contract

| Direction      | Topic / frame                               | Meaning                                                                               |
|----------------|---------------------------------------------|---------------------------------------------------------------------------------------|
| input          | <code>/lio/odom</code>                      | Super-LIO pose and 6×6 covariance for motion uncertainty propagation.                 |
| input          | <code>/lio/cloud_world</code>               | Registered LiDAR cloud, transformed to the map at its timestamp.                      |
| optional input | <code>/camera/.../depth/color/points</code> | D435/D435i cloud; directly fused with the structured-light model.                     |
| output         | <code>/autonomy_light/elevation_map</code>  | Native <code>grid_map_msgs/msg/GridMap</code> , layers: elevation, upper/lower bound, |

|    |                                              |                                                                                   |
|----|----------------------------------------------|-----------------------------------------------------------------------------------|
|    |                                              | uncertainty range.                                                                |
| TF | map → odom → imu → base_link →<br>lidar_link | Super-LIO owns dynamic transforms;<br>launcher owns only calibrated static links. |

The local grid follows `base_link` in `map` with Grid Map's circular buffer. It preserves overlapping posterior cells while moving and clears only newly exposed strips. Unobserved terrain remains `NaN`; no arbitrary startup floor value is inserted.

## Uncertainty and cleanup

Each input uses its own laser or D435 noise model. At each cloud timestamp the elevation node obtains the Super-LIO covariance, predicts map uncertainty due to robot motion, and applies the measurement update in the map frame. Its visibility cleanup ray-traces newer observations to remove penetrated stale cells, addressing lingering wall and overhanging-obstacle artifacts without an additional local implementation.

## Prebuilt terrain input

`elevation_mapping.underlying_map_topic` accepts a prebuilt `GridMap` carrying an `elevation` layer. A Super-LIO PCD remains the map used by `--map` relocation and is not silently reinterpreted as terrain. This keeps localization and terrain-map contracts explicit.

## Operational commands

- `./build.sh` installs/builds Livox, Super-LIO, Grid Map, vendored KINDR/KINDR ROS, the mapper, and bringup.
- `./launch.sh --vis` opens a latest-frame GridMap elevation viewer; `--rviz` opens the GridMap RViz display.
- `./mapping.sh --output maps/site.pcd` runs Super-LIO full SLAM and writes the optimized PCD at shutdown.